

**ADMINISTRATIVE INFORMATION****FMI No: 25371****FMI Title: VCT Tilt Pad Upgrade**

FMI TYPE	CHARGE CODES	IMPLEMENTATION DATES
☐ Mandatory Safety	GEMS - AM:	Issue Date:
☒ Mandatory	GEMS - E:	
☐ Mandatory on Request	GEMS - A:	Close Date:

☐ Staged FMI	Customer List provided by FMI Administration. If you identify additional units within effectivity range, but they are not shown on the customer list, contact FMI Administration in your Pole.
☐	Parts and any special tools are furnished automatically. To address shortages and ordering exceptions, contact FMI Administration in your Pole.
☐	Parts Ordering: GEMS - Americas, Contact your Region logistics coordinator GEMS - Europe, Order FMI kits from GPO/EPL Buc GEMS - Asia, Contact your region logistics coordinator
☐	No Parts are required to complete this FMI.
☐ Batch with	_____
	To minimize costs, combine implementation of this FMI with the FMI's listed for all customers where the effectivity is the same (Please see your customer lists).
☐ Non-Staged	No Customer List is provided. Each service location must identify affected units, and place part orders as indicated in the Special Instructions.

Technical Support		
GEMS - Americas	GEMS - Europe	GEMS - Asia
GE Medical Systems Insite Center 800-321-7937 Milwaukee, WI 53201, USA	GEMS - E / ESC BP34 F 78533 BUC CEDEX, FRANCE Tel: (33 1) 39 20 00 07 or ESC Fax: (33 1) 30 70 99 70	Asia Support Center Phone: 81-426-56-0033 Fax: 81-426-48-2905

Special Instructions:**Unit Tracked:****FMI Instructions Part No.: 5144789-100****Do NOT leave unsecured on site**

FMI Completion Certification Report

Mail To

GEMS - Americas

G.E. Medical Systems
Attn: FMI Admin. W-523
P.O. Box 414
Milwaukee, WI 53201

GEMS - Europe

GEMS - Europe
FMI Admin. 322
BP34
F78533 Buc Cedex, France

Send to your
Country FMI or Service
Administrator

FMI No.: 25371

FMI Title: VCT Tilt Pad Upgrade

Customer Full Name:

Number and Street Address:

City and Country:

Country Code

Site Number

System I.D.

Unit Tracked:

Model Number	Serial Number	Compl. Date DD/MM/YY	FMI Comp. Code	Service Hours	
				Labor	Travel

FMI Completion Code

1. Modification Complete or previously modified.
2. FMI not required per FMI instruction.
3. Not modified. Unit in storage. FMI kit returned to stock.
4. Modification refused or equipment altered. Customer signature and title required.
5. Unit location unknown or unit not in area. Product Locator notified.

Customer Refusal or Equipment Altered - Modification Pending.

Installer
Comments

Effectivity

The following CT Equipment is affected:

Product	Model #
LS7x VCT	5124069 - LS7x VCT Installs Prior to August 1, 2005

Table 1 – Effectivity Table

Purpose

This FMI incorporates three Sections.

1. Replace the front gantry cover tilt pad sensor and rework cable restraint. Some M3 gantries have cable restraints that could possibly come in contact with the Collimator. Reworking the cable restraints and adding a new tilt pad with a longer lead will eliminate failures due to cables being damaged by the rotating assembly.
2. Replace the 48v PS Guard on the stationary portion of the Gantry with a guard that has ventilation slots.
3. Replace the Dynamic Brake assembly 7 Amp fuse with an 8 Amp fuse.

Furnished Materials

Qty	Description	Part #
<u>Tilt Pad Sensor Sec. 1</u>		
1	FMI Document	5144789-100
1	3M DP100 Structural Adhesive	5130489
1	9740 Epx Man Gun 50ML 3M Duo-Pak Applicator – Manual Epoxy Gun	2298033
2	Mix Nozzle Devcon Mark 5 System for Use with 3M Manual Epoxy Gun – Static Mixer Nozzle (One used in this FMI. One is extra)	5147983
1	SANDPAPER,320 GRIT,WET/DRY	2052087
1	TEXWIPE 58019-506-91/9% ALCOHOL PADS	46-183039P1
2	Adhesive Mount – Cable Tie (One used in this FMI. One is extra)	46-208747P2
2	Cable retainer (One used in this FMI. One is extra)	5136449
3	11.08~X .140~ Self-locking Cable Tie (One used in this FMI. Two are extra)	46-208758P9
1	Sensor, Patient Tilt, Small, 175mm long tail with bend for low profile cover routing	5134416
2	Washer, 4.3MM ID X 20MM OD X 1MM Thick, Bossard BN10342	5159601
2	NUT, M4 X 0.7 HEX WITH LOCK WASHER ZINC PLATED	46-312346P2

Qty	Description	Part #
<u>Power Supply Guard Sec. 2</u>		
1	Guard,48v PS,VCT	5128606
1	SCREW HEXAGON SOCKET 4 MM 10 MMZINC PLATED, STEEL	46-328417P3
1	Washer Lock – Spring 4.1 MM 7.6 MMZINC Plated, Spring Steel	46-328432P2
1	Washer Plain – Normal 4.3 MM 9 MMZINC Plated, Steel	46-328430P2
<u>Fuse for Dynamic Brake Sec 3</u>		
1	FUSE 8 AMPS 250 VOLTS BUSSMANN ONLY	46-170021P106

Tools Required

Qty	Description	SW Version	Part #
1	Flat Blade Screwdriver		FE Tool
1	Side Cutter		FE Tool
1	7MM Socket with driver		FE Tool
1	Pair of latex or vinyl gloves		FE tool
1	Flexible tape measure in millimeters		FE Tool
5	Ty-raps		FE Tool

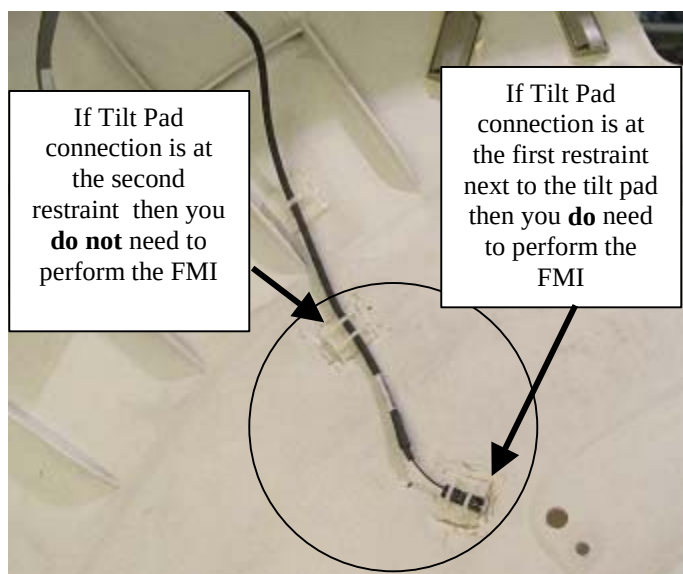
Personnel Requirements

"3.0" hours labor and "1" hour of travel

Procedure

Preparation

1. Reference picture below to ensure you need to perform FMI
2. Inventory FMI kit
3. Disable gantry power using the Service Switches, 120 VAC, Axial Enable, HVDC.
4. Perform proper Lockout/Tagout power control procedures
5. Remove Gantry Front and Rear Covers.



Determine need for FMI by identifying Tilt Pad connection point

Hardware Procedure

Follow the steps listed below:

Section 1 – Tilt Pad Sensor

1. Remove gantry front cover and identify the two cable restraints to be replaced. See Figure 1.

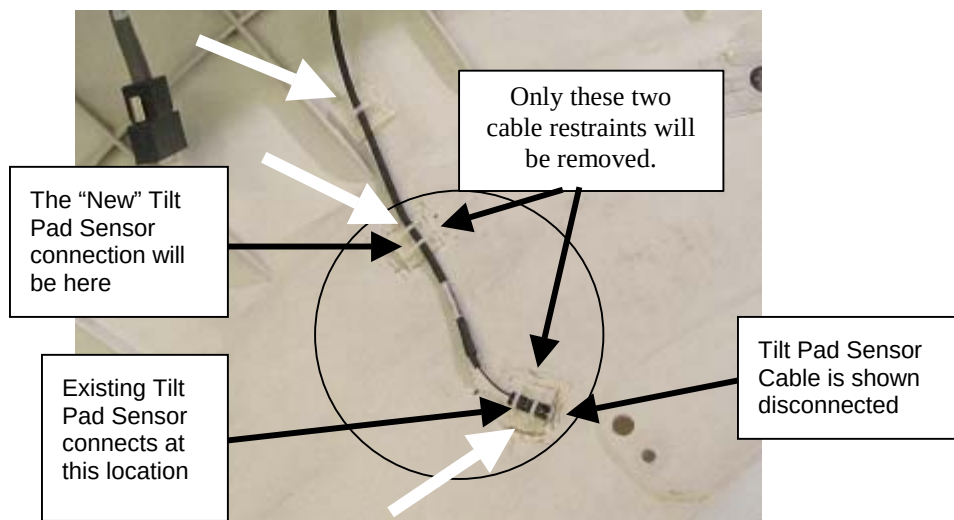


Figure 1 Identify Cable Restraints to be Removed

2. Disconnect the Tilt Pad Sensor Cable and carefully cut the Ty-raps, in three places as indicated with white arrows, that hold the cable in place.
3. Use a flat tipped screwdriver to carefully pry (remove) the two lower cable restraints from the cover. See Figure 2

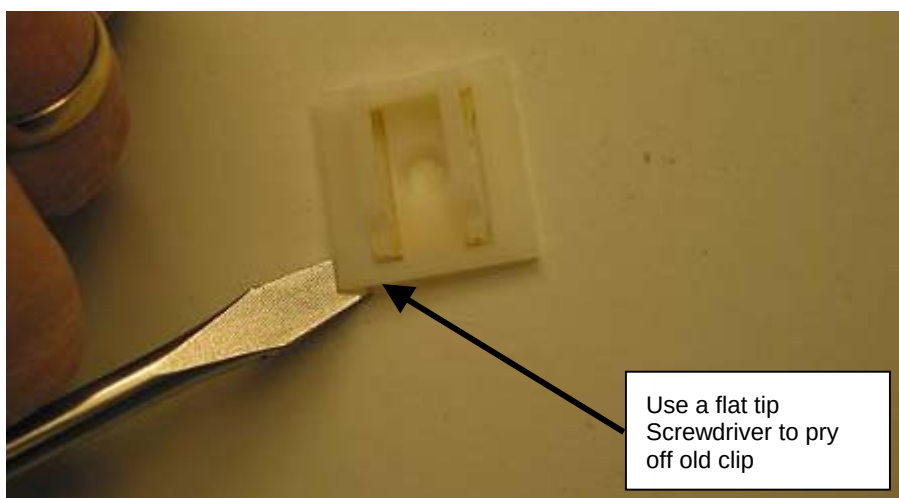


Figure 2 Remove old clip.

4. Covers manufactured by Yashiro Plastics will be painted white on the inside and have indents locating where the old pads and new pads are to be placed. Scrape the old glue off with screwdriver and using sandpaper, sand off the paint where the new clips will be placed. Clips will not stick to paint!
5. If your cover is manufactured by Parkway Plastics, it will be brown in molded color and have no indents and you will need to measure the location of the new clips. Reference Figure 3 for location of new clips.

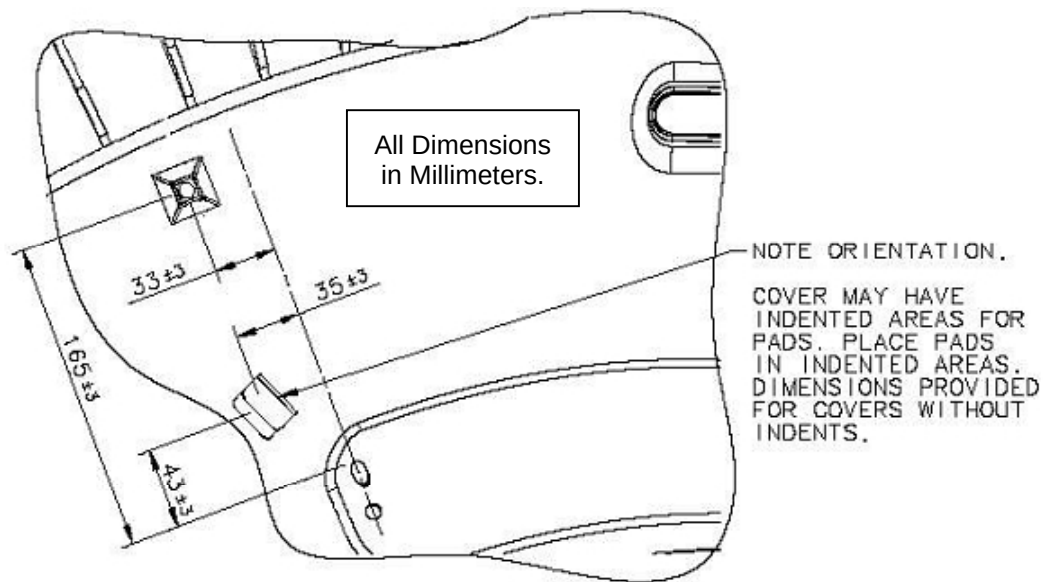


Figure 3 New Restraint Locations and Orientation

Clean off the area with alcohol pads (to remove dust) but don't throw the pads away yet. The front gantry cover (manufactured by Yashiro) will now be brown in color when properly sanded. See Figure .



Figure 4 Cable Restraints Removed

6. Prepare the new cable restraints as follows:
 - a. Take 46-208747P2 (Adhesive Mount Restraint) and remove all adhesive and clean with alcohol pad. See Figure 4
 - b. Use sandpaper to roughen bottom surface on both restraints for better adhesion.

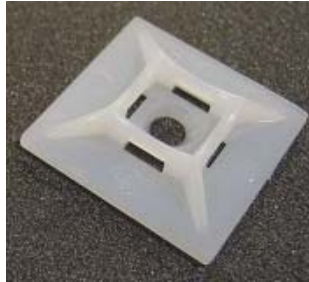


Figure 3 46-208747P2



Figure 4 5136449

7. Assemble the epoxy gun, epoxy, and mixer using the 1:1 plunger insert. Two mixers are provided, (one is a spare). **Note position of the opening of the clip restraints.** . With the 3M DP100, attach the clips into the indents on the newer covers and on the older covers as shown in Figure . Just use enough epoxy to put a thin film on back of the clips. Too much epoxy will fill in the ty-rap slots and also cause the clips to float down the cover until the epoxy sets up. It will take 5 minutes to set. Note: The epoxy will set quickly inside the gun so use immediately after opening.

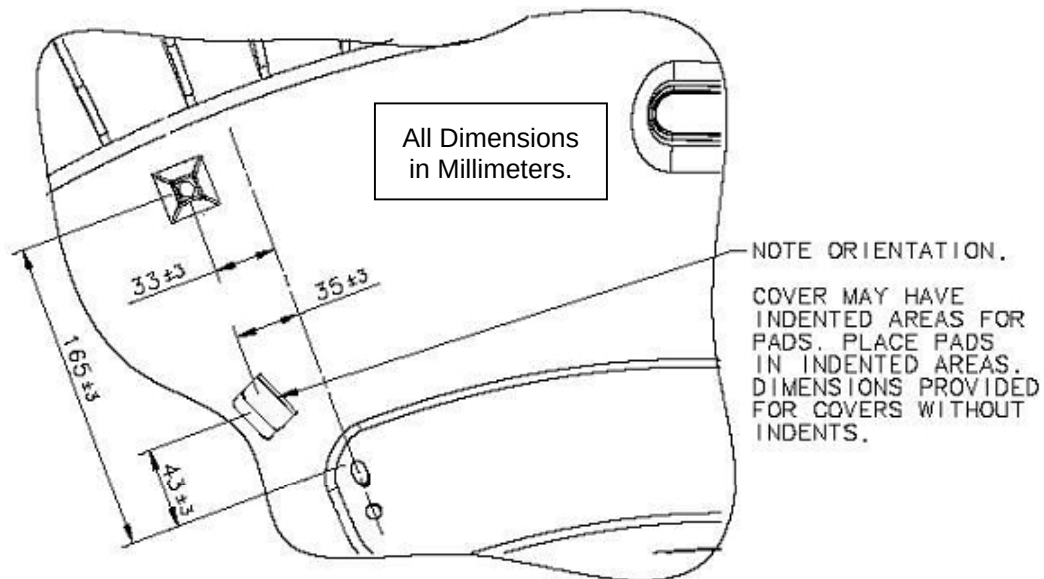


Figure 5 New Restraint Locations and Orientation

8. Install the tilt pad harness. Add Ty-raps as shown in Figure 7 and trim excess ty-rap material.

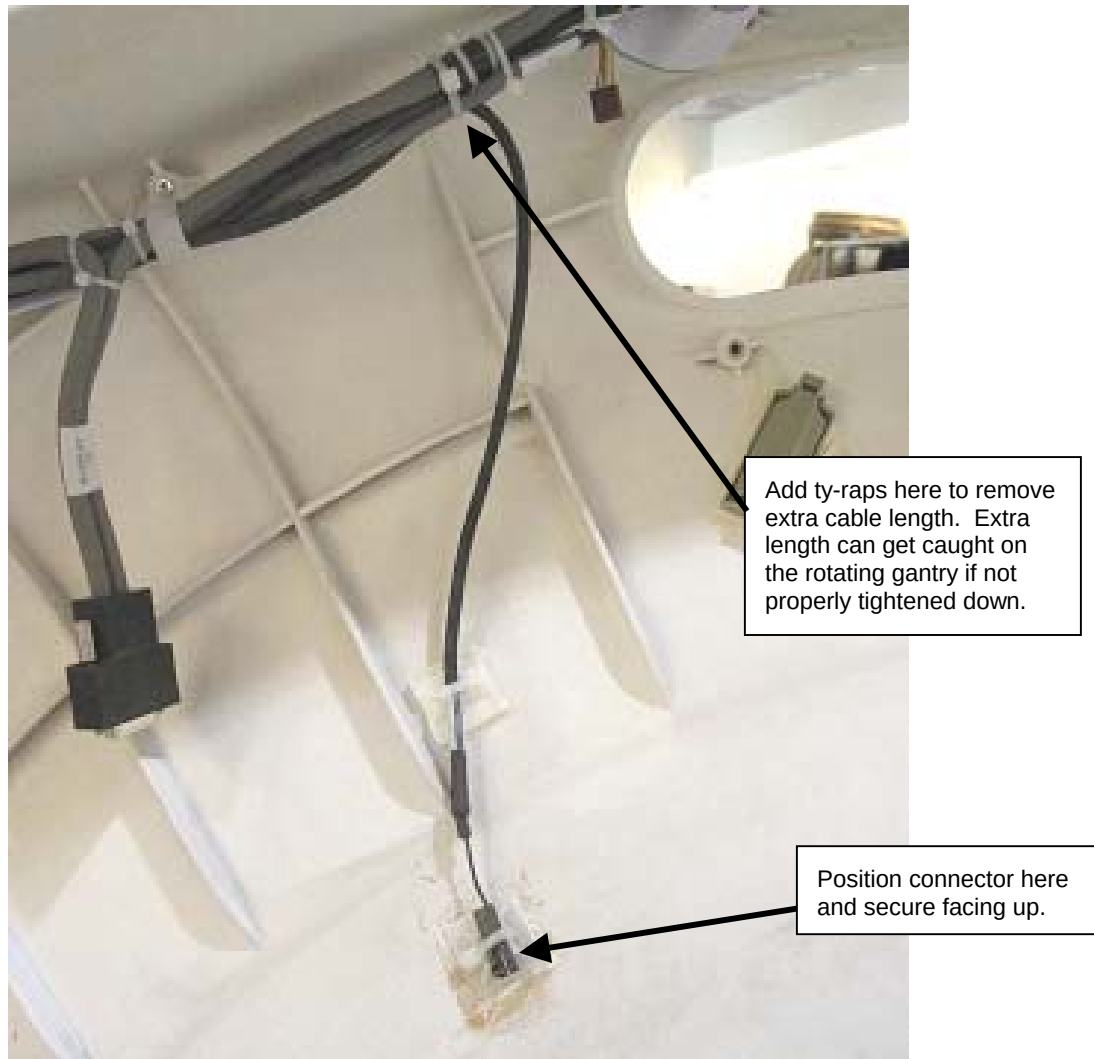


Figure 7 Front Cover Cable Ty-rap

9. Remove Tilt Sensor Pad by removing two nut fasteners and replace with new Tilt Pad Sensor. See Figures 8 & 9.

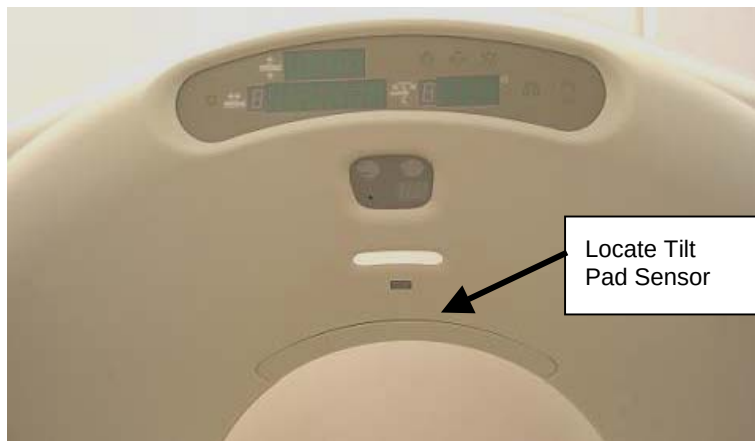


Figure 8 Tilt Pad Sensor Location

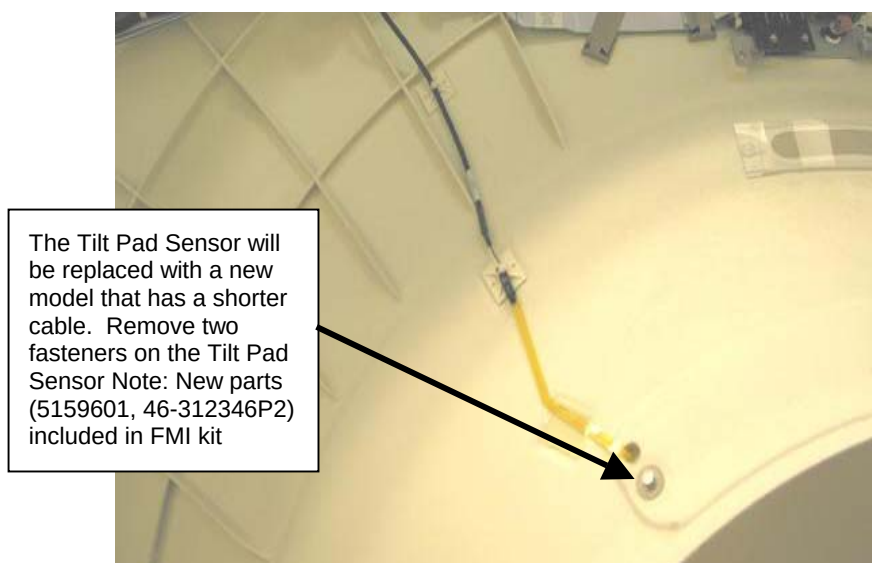


Figure 9 Tilt Pad Fasteners

10. Orient the harness within the restraint as shown in Figure 10. When assembled, it should look like the picture in Figure 10.
 - a. 5134416 New Tilt Pad Tail Connector will connect facing up.
 - b. 5134416 New Tilt Pad Tail slides into clip 5136449.
 - c. Verify that all extra material is removed...adjust as required.

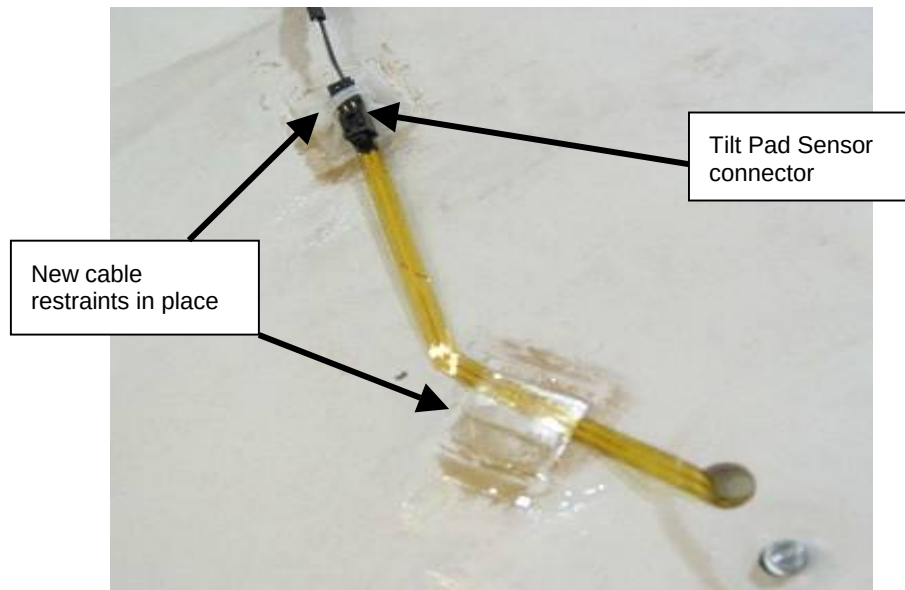


Figure 10 New Tilt Pad Sensor Harness Routing

11. Ty-rap connector facing up to the 46-208747P2 restraint. See Figure 11.

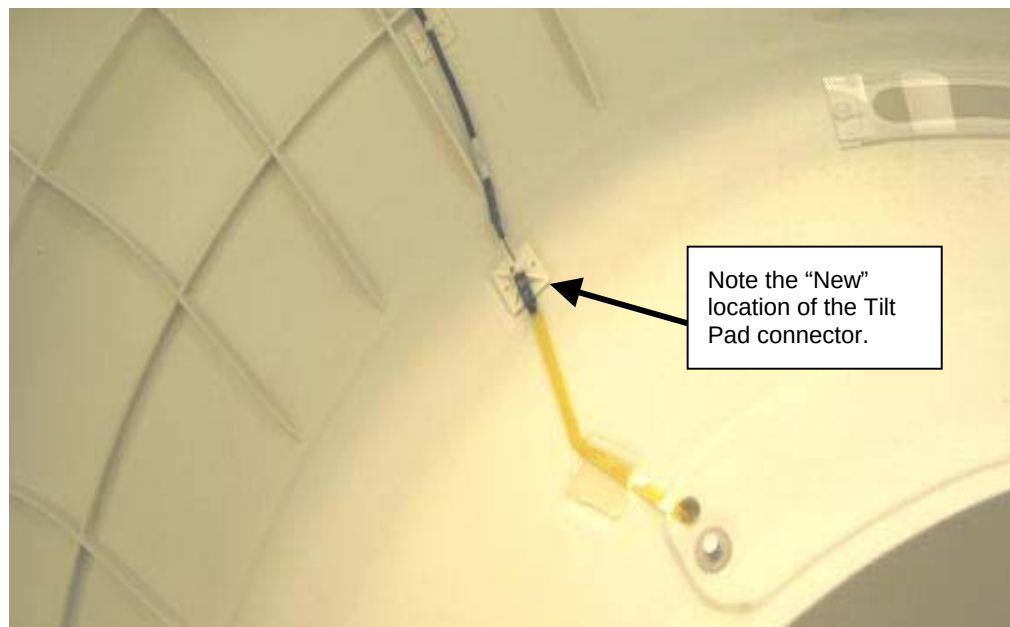


Figure 11 Front Cover Cable Clearance

This concludes the Tilt Pad modifications. Please move on to Section 2. "48v Power Supply Guard"

Section 2 – 48V Power Supply Guard

The 48v Power Supply has a plastic guard that must have ventilation slots as shown in Figure 11. Remove rear gantry cover. If the system has the guard without slots then remove the guard by loosening the screw fastener shown in Figure 12. Replace with new guard supplied in this kit.

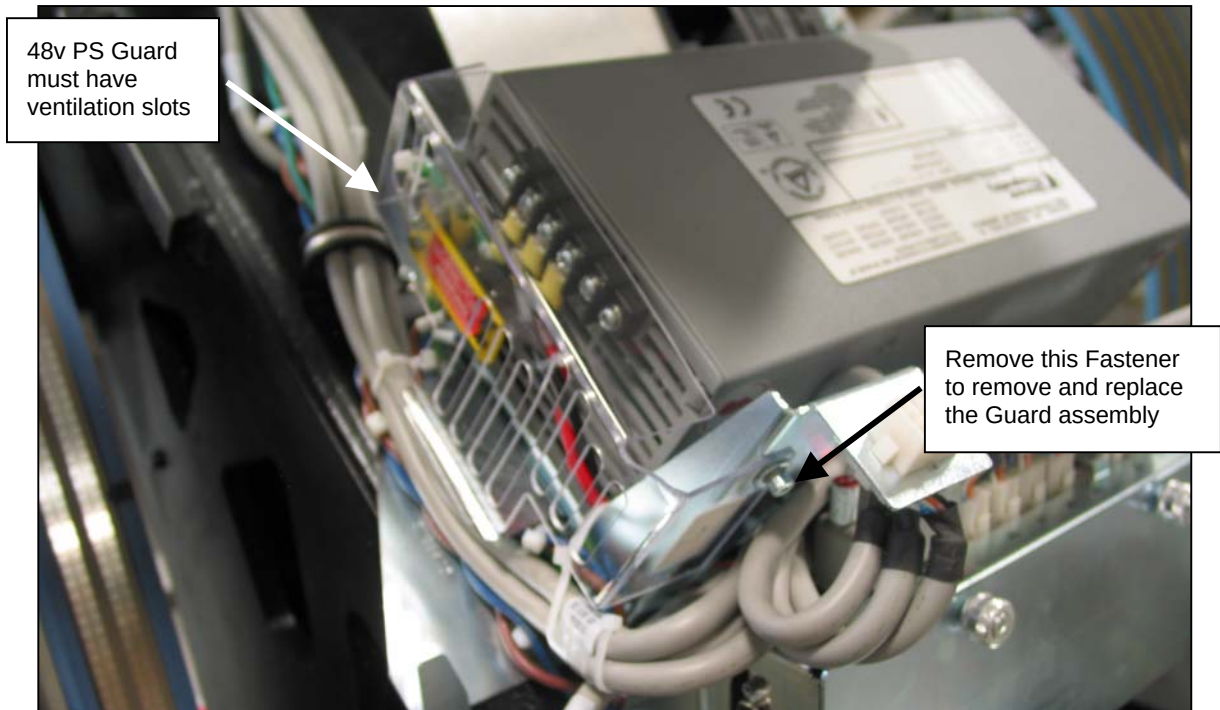


Figure 12 New Tilt Pad Sensor Harness Routing

Section 3 – 8 amp Fuse Replacement

The fuse located below the Dynamic Brake assembly needs to be rated at 8 AMPS. Remove the existing 7 AMP fuse and replace with the 8 AMP fuse supplied in this kit. See Figure 13.

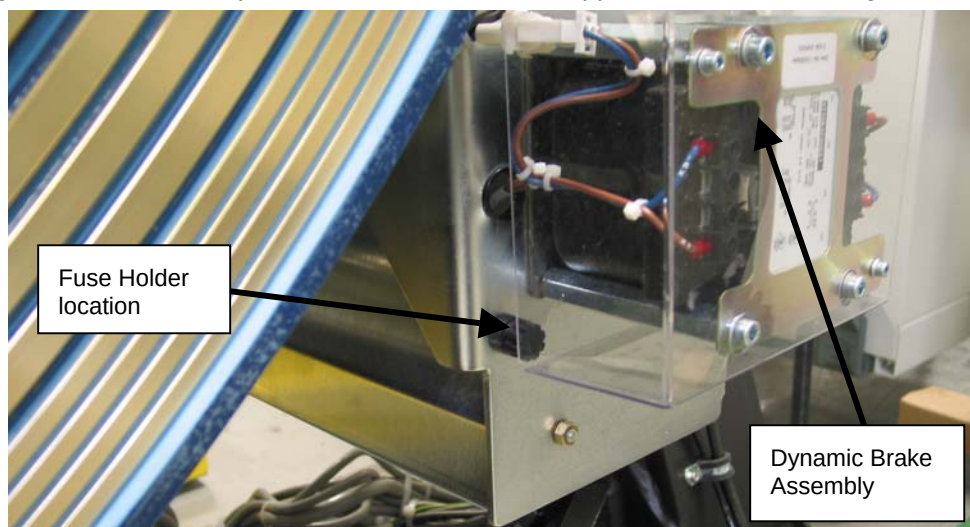


Figure 13 8 AMP Fuse Location

Section 4 - Finishing Up

1. Restore gantry power, 120 VAC
Rotate gantry by hand and observe as best you can to ensure harness clearance.
2. Turn on Axial Enable and HVDC.
3. Install all gantry covers.
4. Tilt gantry and press on new tilt pad sensor. Gantry tilt should stop.
5. Perform test scans to verify system operation.

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